

Raast Integrated Digital Wallet and Transaction Monitoring System



Course: CS2013 Introduction to Database Systems

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A. Project Description

This project focuses on the design and implementation of a real world fintech database system centered around a digital wallet platform integrated with Pakistan's Raast payment infrastructure. The system is intended to simulate how modern mobile wallets operate by allowing users to securely store funds, perform peer to peer transactions, and make payments to registered merchants.

The proposed database system will manage multiple core components including user accounts, wallet balances, transaction records, merchant payments, and linked bank accounts. It will ensure data integrity, accuracy, and consistency by applying relational database principles such as normalization, constraints, and structured query processing. Additionally, the system will support real time transaction recording and balance updates, reflecting the behavior of actual fintech platforms.

A key feature of this project is the inclusion of transaction monitoring and fraud detection mechanisms. The system will identify suspicious activities such as high value transfers or repeated failed transactions and flag them for further analysis. This enhances the reliability and security of the system.

Overall, this project demonstrates the practical application of database concepts in a rapidly growing fintech environment and highlights how structured data systems support secure and efficient digital financial services.

Problem Statement

With the rapid shift towards digital financial services, systems handling monetary transactions must ensure high levels of accuracy, security, and efficiency. Digital wallets and instant payment platforms process thousands of transactions daily, involving sensitive user data, financial balances, and real-time updates. Without a well-structured database system, managing such operations becomes highly inefficient, leading to data inconsistencies, duplication, transaction errors, and increased risk of fraud.

Traditional or poorly designed systems may fail to maintain data integrity, resulting in incorrect balances, lost transaction records, or unauthorized access. Moreover, the absence of proper monitoring mechanisms makes it difficult to detect suspicious activities such as unusually large transactions or repeated failed attempts.

Therefore, there is a strong need for a robust database system that can efficiently store, manage, and process financial data while ensuring consistency, reliability, and security.

This project aims to address these challenges by designing a structured and scalable fintech database solution.

Motivation

The increasing adoption of digital payment systems in Pakistan has significantly transformed the financial landscape. With platforms such as mobile wallets and instant payment systems becoming widely accessible, users now rely heavily on digital transactions for everyday activities such as bill payments, online shopping, and peer-to-peer transfers.

This rapid growth has created a demand for efficient backend systems capable of handling large volumes of transactional data. The motivation behind this project is to develop a database system that reflects real-world fintech operations and demonstrates how structured data management can support secure and efficient financial services.

Additionally, this project provides an opportunity to apply theoretical database concepts such as normalization, relational design, constraints, and query optimization in a practical environment. By integrating features such as fraud detection and transaction monitoring, the system also highlights the importance of security and analytics in modern financial technologies.

Real World Relevance

The proposed system closely reflects real-world fintech platforms such as Easypaisa, JazzCash, and Raast. These platforms allow users to store money digitally, perform instant transactions, and interact with merchants in a secure environment.

By modeling similar functionalities, this project demonstrates how database systems are used in real financial applications to manage user accounts, transaction histories, and payment processing. It also reflects industry practices such as KYC verification, balance tracking, and fraud detection.

Understanding such systems is highly relevant for students pursuing careers in fintech, banking, and data management, as it provides insight into how modern financial infrastructures operate behind the scenes.

B. System Modules

Module 1: User and KYC Management

This module is responsible for managing user registration and verification processes. It stores essential user information such as full name, CNIC, phone number, email, and address. Additionally, it maintains the KYC status of each user, ensuring that only verified users can access full wallet functionalities.

The module also enforces constraints such as unique CNIC and phone number to prevent duplicate accounts. It supports operations such as user registration, profile updates, and verification status changes.

Example:

A new user registers using their CNIC and phone number. The system verifies the details and updates the KYC status from pending to verified before allowing transactions.

Module 2: Wallet Management

This module manages digital wallets associated with users. It stores wallet details such as wallet number, balance, wallet type, and status. The system ensures that each user has a unique wallet and maintains accurate balance updates after every transaction.

It also handles conditions such as blocking inactive wallets or restricting transactions when the balance is insufficient.

Example:

When a user receives money, their wallet balance is automatically increased. When they make a payment, the balance is reduced accordingly.

Module 3: Transaction Management

This module is the core of the system and handles all financial transactions. It records details such as sender wallet, receiver wallet, transaction amount, type, date, and status. It supports various transaction types including wallet-to-wallet transfers and merchant payments.

The module ensures transactional integrity by maintaining accurate records and preventing invalid operations such as transferring more money than the available balance.

Example:

User A sends Rs. 5000 to User B. The system records the transaction, updates both wallet balances, and marks the transaction as successful.

Module 4: Fraud Monitoring and Reporting

This module enhances system security by identifying and managing suspicious activities. It monitors transaction patterns and flags unusual behavior such as high-value transactions or repeated failures.

The system stores fraud alerts along with reasons and severity levels, enabling further investigation.

Example:

If a transaction exceeds Rs. 100,000, the system automatically flags it as suspicious and records it in the fraud alert table.

C. Database Design

Entities

The system consists of several key entities including User, Wallet, Transaction, Merchant, Bank Account, and Fraud Alert. Each entity represents a real-world object and contains attributes relevant to its function.

For example, the User entity stores personal and identification details, while the Transaction entity records financial operations.

Relationships

The database is designed using relational principles to ensure proper data organization and integrity. A user can have one or more wallets, while each wallet is associated with a single user. A wallet can perform multiple transactions, either as a sender or receiver.

Transactions may also be linked to merchants in case of payments, and certain transactions can generate fraud alerts. These relationships ensure a structured flow of data and enable efficient query processing.

D. Functional Requirements

The system must support user registration and maintain unique identification records. It should allow the creation and management of digital wallets linked to users. The system must enable secure financial transactions, including transfers and payments, while ensuring sufficient balance and valid transaction conditions.

Additionally, the system should detect and flag suspicious activities, maintain transaction history, and generate meaningful reports. It must also enforce data integrity through constraints and relationships, ensuring consistency across all operations.

E. Queries and Reports

The system will support a variety of SQL queries to analyze and retrieve data. These include retrieving transaction history for specific users, calculating total transaction amounts, identifying failed or suspicious transactions, and generating monthly summaries using aggregation functions such as GROUP BY.

Reports will provide insights into user activity, transaction trends, and system performance. These queries demonstrate the practical use of joins, subqueries, and aggregation functions in real-world scenarios.

F. Tools and Technologies

The project will be implemented using MySQL as the primary database management system. Microsoft Access will be used to design forms and reports for user interaction and data visualization. Draw.io or similar tools will be used to create ER diagrams and illustrate database structure.

Additional tools such as Microsoft Excel may be used for managing sample data, while documentation will be prepared using Microsoft Word or similar software.

G. Advanced Features

The system includes several advanced features to enhance functionality and performance. Triggers will be used to automatically update wallet balances after transactions. Views will be created to simplify complex queries and provide summarized data.

Indexing will be applied to frequently searched fields such as CNIC and transaction date to improve query performance. Additionally, basic fraud detection logic will be implemented to identify suspicious activities and improve system security.